Overall Aero Package

NFR25
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Abstract

The NFR25 aero package comprises a front wing, rear wing and undertray for performance gains in cornering. Major focuses included rule compliance, simplified manufacturability and vehicle dynamics integration. CFD Simulations were run using Ansys Fluent and on-track testing was conducted to validate the results. The package was focus in the overall team goal:

"Optimize for autocross while being able to complete endurance"

Justification for Aero

We at Northwestern Racing believe that the fastest cars absolutely require aero. Looking at the top cars from competition year in and year out and over 90% have aerodynamic packages. Simply put, the added benefit of downforce outweighs the negative effects of added mass and added size. Furthermore, the added benefit of downforce outweighs the increase in drag. For electric vehicles, this is very significant since batteries are not energy dense. We ran a spread of Cl and Cd, lift and drag coefficients, through OptimumLap to compare the effects on lap times (see Figure 1). We found that lap times increase nearly linearly with improved Cl/Cd ratios. However, we did realize that the increase in drag heavily impacted our ability to finish endurance. Therefore minimizing the drag and weight experienced by the car was very important. An acceptable downforce to mass ratio was 1.5.

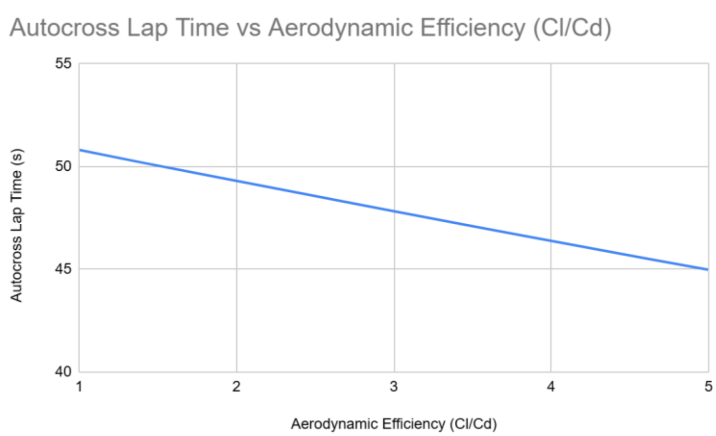

Figure 1: Effect of different values of Aerodynamic Efficiency on the Autocross lap time.

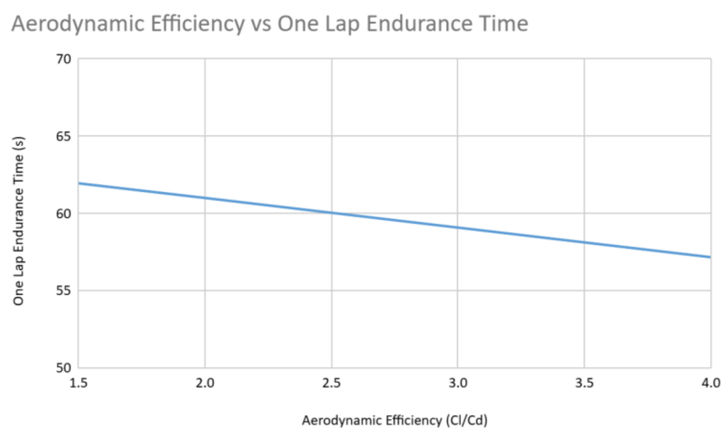

Figure 2: OptimumLap simulation of Aerodynamic Efficiency over 1 lap of Endurance time

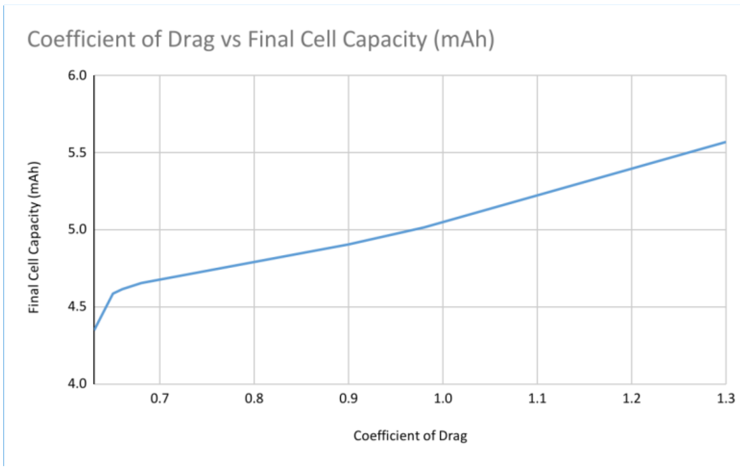

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Figure 3: OptimumLap simulation of Final Cell Capacity against different Coefficients of Drag

Goals

In order to set specific and measurable goals for the aerodynamic package, different values of coefficients of drag, coefficients of lift were iterated over in OptimumLap based on the 2024 FSAE Michigan Autocross and Endurance tracks. Table 1 shows the data accumulated from the OptimumLap simulation.

Coefficient of Drag	Coefficient of Lift	Aerodynamic Efficiency	Total Time	Lap Time	Final Cell Capacity	Maximum Cell Current
0.63	-0.29	-0.46	1433.67	65.167	3.9908	26.83
0.65	0.79	1.22	1405.44	63.884	4.0452	27.12
0.66	1.28	1.94	1392.41	63.291	4.0834	27.34
0.68	1.56	2.29	1385.331	62.97	4.1268	27.59
0.9	2.34	2.6	1369.092	62.231	4.2962	28.73
0.98	2.58	2.63	1364.398	62.018	4.3688	29.49
1.3	3.7	2.85	1341.764	60.989	4.8141	36.09
1.63	1.63	1	1400.148	63.643	4.8114	36.53
1.63	2.445	1.5	1379.802	62.718	4.9235	39.47
1.63	3.26	2	1359.551	61.798	5.082	52.11
1.63	4.075	2.5	1338.772	60.853	5.27	77.21
1.63	4.89	3	1318.395	59.927	5.3889	119.44
1.63	5.705	3.5	1297.555	58.98	5.8488	13632.01
1.63	6.52	4	1276.354	58.016	5.9141	18407
0.63	1.89	3	1376.028	62.547	4.0962	27.41
1.13	3.39	3	1346.44	61.202	4.5859	32.86
1.63	4.89	3	1318.395	59.927	5.3889	119.44
2.13	6.39	3	1292.777	58.763	5.6829	30600.91
2.63	7.89	3	1280.099	58.186	6.525	105089.6
0.63	1.89	3	1376.028	62.547	4.0962	27.41
0.73	2.19	3	1369.973	62.271	4.1694	27.86
0.83	2.49	3	1364.033	62.001	4.2638	28.49
0.93	2.79	3	1358.142	61.734	4.3528	29.31
1.03	3.09	3	1352.278	61.467	4.4523	30.49

A higher final cell capacity higher than 4.5 would result in us being unable to finish endurance therefore we chose the Cl and Cd that gave us a value around 4.5. Considering that OptimumLap models the vehicle as point therefore disregarding vehicle load transfer when driving the car on a track, we anticipate that the final cell capacity will actually be lower than what we simulated from OptimumLap.

From the table above, for endurance, the whole car:

- Minimum Lift Coefficient: -3.39
- Maximum Drag Coefficient: 1.13
- Minimum Downforce: 484 N
- Maximum Drag: 161 N

Taking air density $\rho = 1.293\ kg/m^3$, frontal area $A = 1.23\ m^2$, air velocity $v = 13.4\ m/s$:

$$L = \frac{1}{2}\rho A C_L v^2 = 484\ N$$
$$D = \frac{1}{2}\rho A C_D v^2 = 161\ N$$

Through vehicle dynamic MATLAB code with integrating with suspension, the ideal center of pressure location was determined. This was based on the point where we get the highest maximum lateral acceleration and the stability index at maximum lateral acceleration is zero (0).

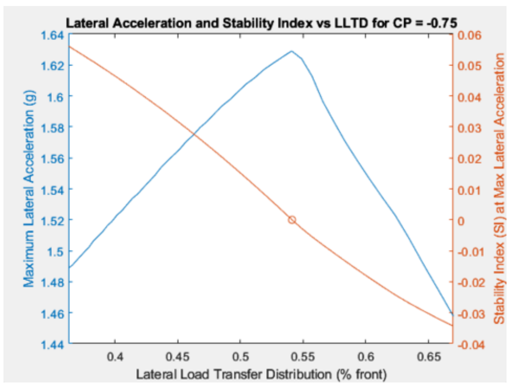


Figure 4: Plot of Maximum Lateral Acceleration, Stability Index and Lateral Load Transfer Distribution

The ideal center of pressure of location was determined to be 0.75” behind the center of the wheelbase. Therefore, it is in front of the center of gravity (2.4” behind the center of the wheelbase)

Design Overview

Previously, the team has always run into issues with aerodynamics boundaries outlined by the rules book. To avoid that this year, conservative offset of 0.75” were set for each project from each bounding plane of the aerodynamic boxes.

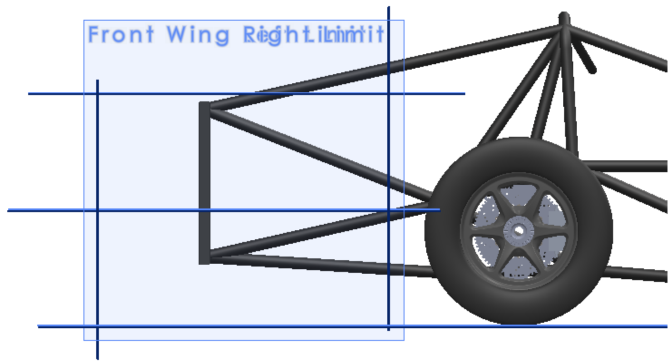


Figure 5: Front Wing bounds side view

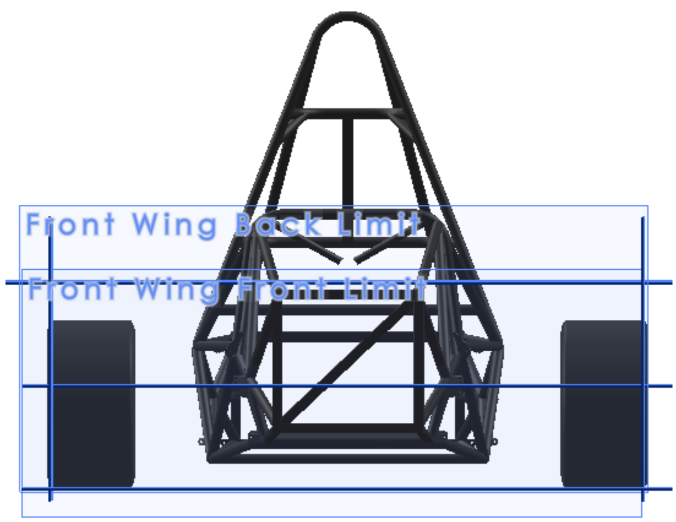


Figure 6: Front Wing bounds top view

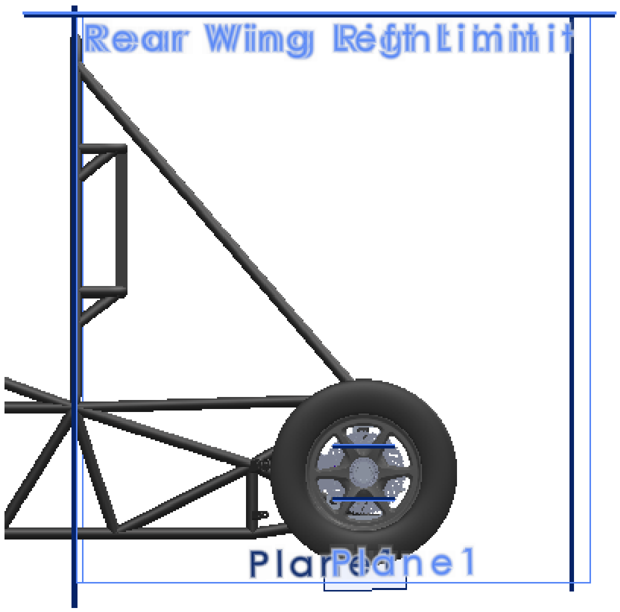


Figure 7: Rear Wing Bounds side view

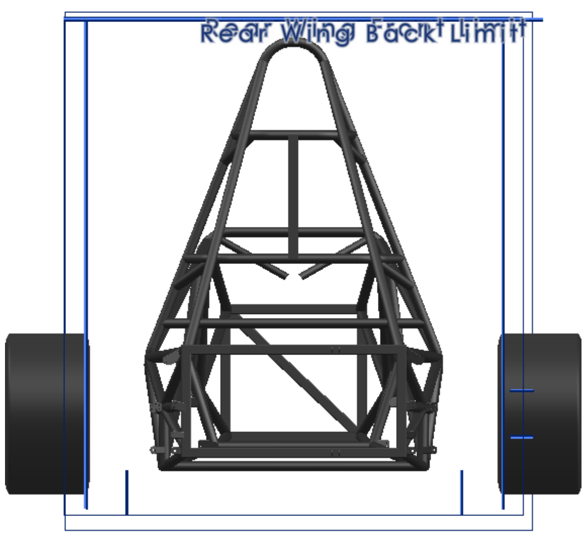


Figure 8: Rear Wing bounds back view

Airfoil selection

The first step in designing the package was selecting airfoils for the front and rear wing. The aim was to find high lift - low Reynolds number airfoils and choose the airfoil with the highest L/D ratio and high lift. Different airfoils were picked from the airfoil database and 2D simulations were conducted.

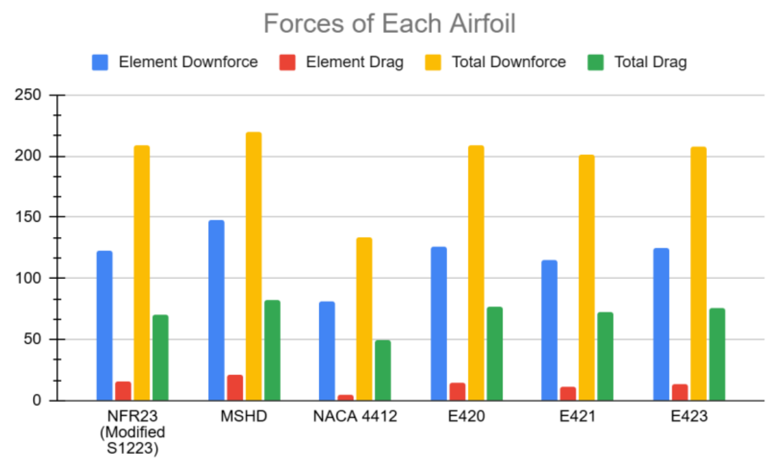


Figure 9: Downforce and Drag in Newtons of high lift airfoils

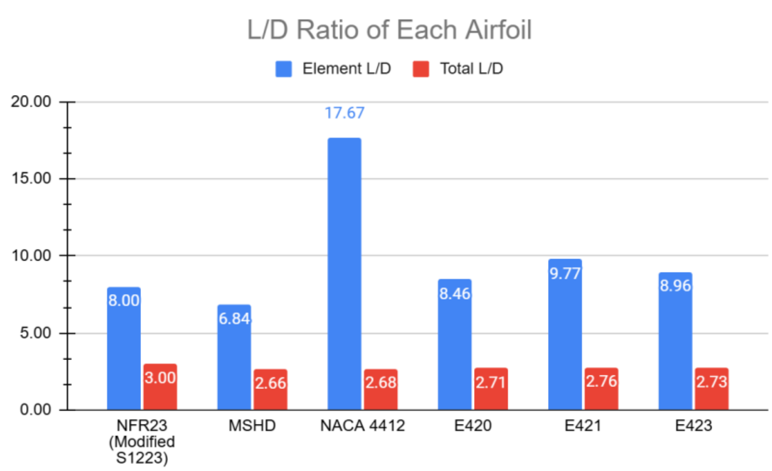


Figure 10: L/D ratio of different airfoils.

From Figure 9 and 10, the E423 airfoil was chosen for all the rear wing elements because it provided a considerable L/D ratio and high lift compared to the other airfoils. Even though the NACA 4412 had a very high L/D ratio, its downforce values were pretty low compared to other airfoils.

The front wing experiences ground effect from its proximity to the ground. Therefore, we psearched for symmetric airfoils for the front wing main foil which allowed for more predictable behavior in ground effect. This is because symmetrical airfoils have zero camber and hence create a more even pressure gradient which helps maintain attached flow under tight ride height conditions and reduce flow separation near the leading edge. Ultimately, we settled on the **NACA 0012.** The second and third element airfoils were developed from the **S1223** airfoil due to its high lift capabilities.

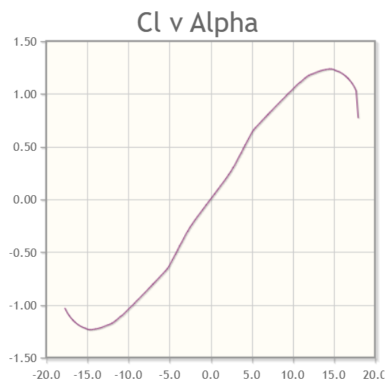


Figure 11: Cl vs Alpha plot of the NACA 0012 airfoil

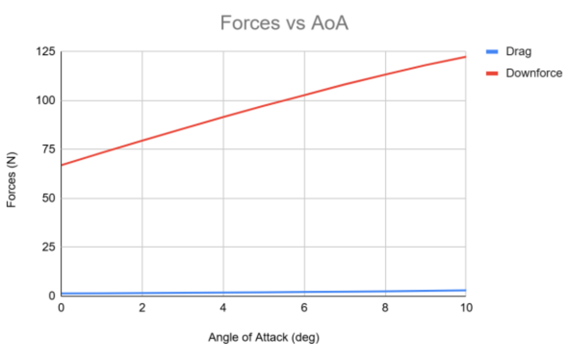


Figure 12: Downforce and Drag in Newton's comparison of the S1223 airfoil.

Front wing Airfoil Data:

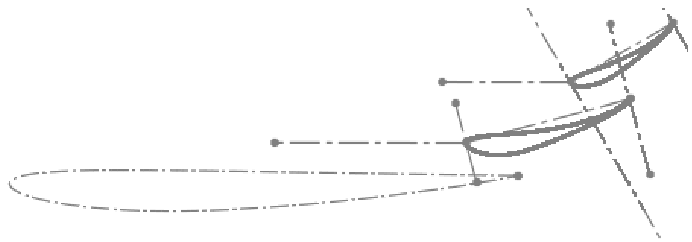


Figure 13: Element outline of the front wing

- Main foil (NACA 0012): chord length 15”, 100% thickness, 5 deg AoA
- Second and third foil (S1223): chord length 5“ & 3.5”, 100% thickness, AoA 15 deg & 30 deg

Rear Wing Airfoil Data:

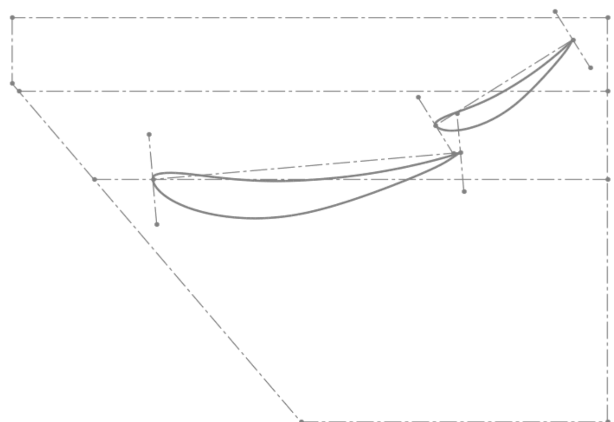


Figure 14: Element outline of the front wing

- Main foil (E423): chord length 21”, 100% thickness, AoA 5 deg
- Second foil (E423): chord length 11”, 100% thickness, AoA 32 deg

<u>Multi-element Wings</u>

We chose to make the front wing and rear wing out of multiple elements because they are able to generate more downforce at higher speeds with minimal flow separation compared to using one whole single element. Three elements were chosen for both the front wing and two elements for the rear wing. Active aerodynamics would be incorporated in the rear wing in order to reduce in straights.

<u>Component Design</u>

Front Wing

The front wing serves as the main aerodynamic surface in the front of the car and directs airflow over other components of the aero package. It provides downforce for better cornering performance and creates clean air for other wings/foils.

Design Goals:

- Provide downforce in front of the car
- Follow aero limits
- Create a simple and compact wing
    - Reduce number of elements from last year
    - Simple rigid mounting

CAD:

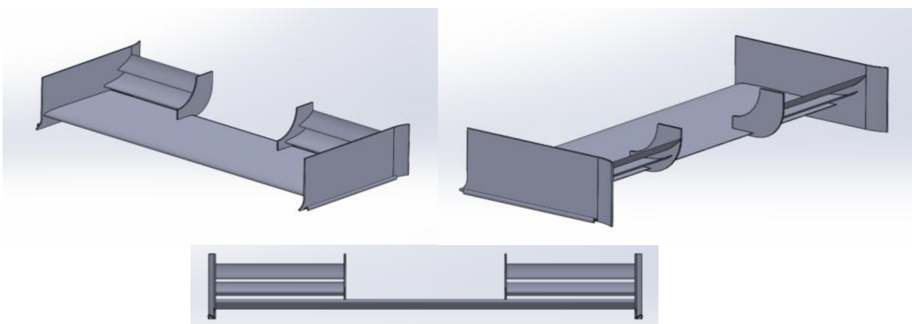


Figure 15: Multiple views of the front wing

CFD @ 30 mph:

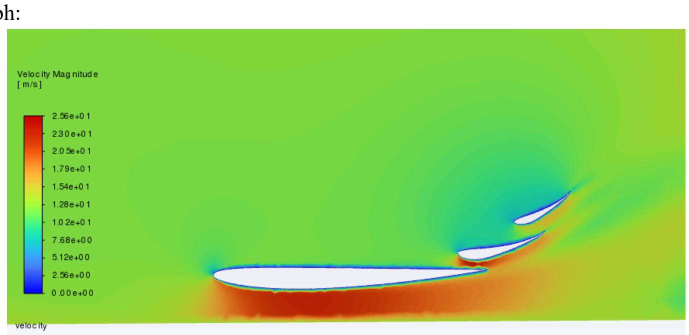


Figure 16: 2D Velocity plot of the side front wing elements at 30 mph

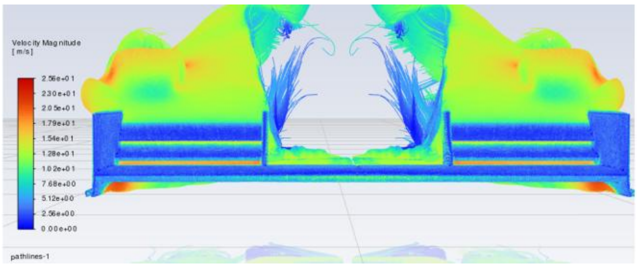


Figure 17: 3D Velocity plot of the front wing elements at 30 mph

Result @ 30 mph:
- Lift: -122.4N
- Drag: 10.24 N
- L/D ratio: 12

Rear Wing

The rear wing serves as the main aerodynamic surface in the rear of the car. The resultant downforce helps the car tires grip the ground, which allows for faster cornering, which results in faster lap times

Design Goals:
- Keep it simple
- More stable and rigid mounting
- Provide downforce on the rear axle of the car
- Get working design as soon as possible

CAD:

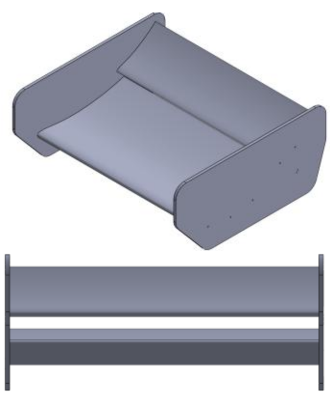


Figure 18: Multiple views of the rear wing

CFD @ 30 mph:

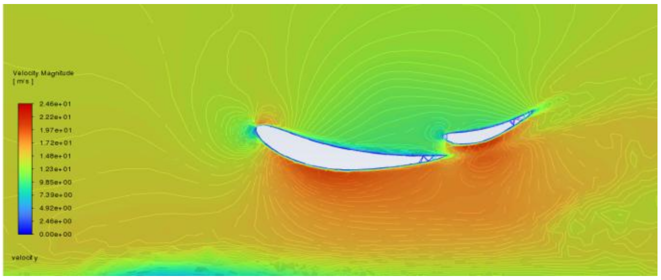


Figure 19: 2D Velocity plot of the side rear wing elements at 30 mph

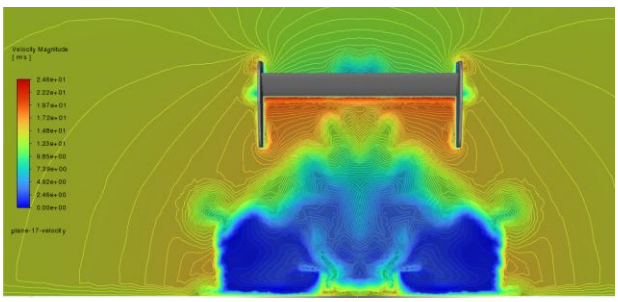


Figure 20: 2D Velocity plot of the back rear wing elements at 30 mph

Result @ 30 mph:

- Lift: -108N
- Drag: 26 N
- L/D ratio: 4

<u>Active Aero</u>

At 20% throttle input, the active aerodynamics on the rear wing will be open and the angle of attack or the second element changes from 32 deg to 10 deg.

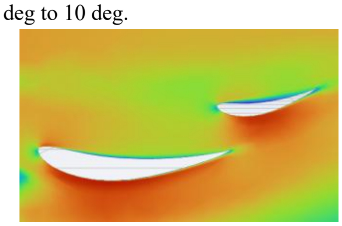


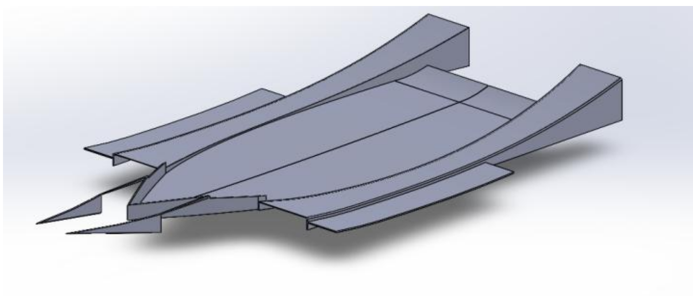
Figure 21: Velocity plot when active aero is on

Results:
- No DRS: Drag 388 N
- DRS: Drag 317 N
- DRS Effectiveness: 17% reduction in drag

Undertray

Design Goals:
- Produce maximum downforce
- Guide turbulent air away from rear of car

CAD:



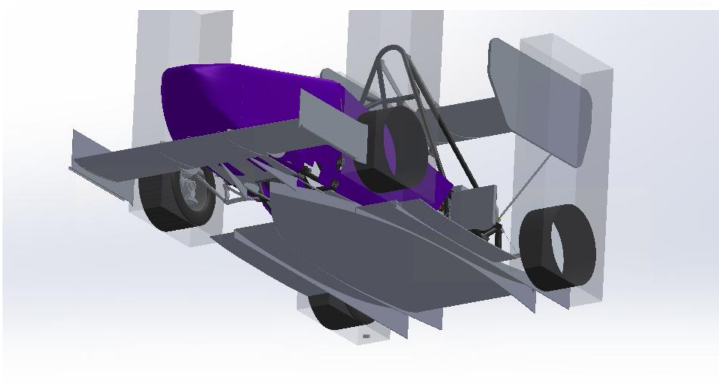


Figure 22-23: CAD images of the undertray

CFD:

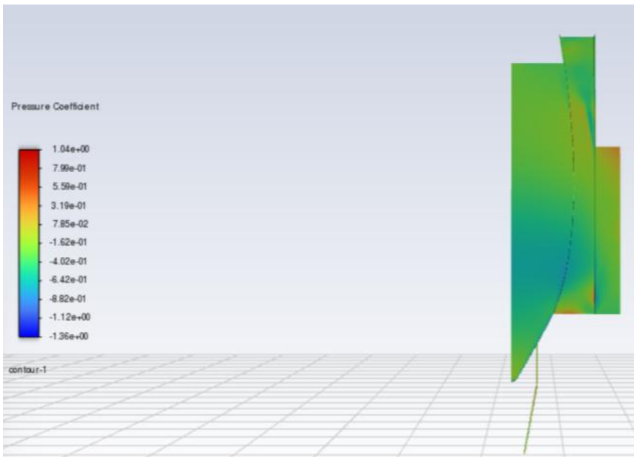


Figure 24: Pressure Coefficient surface plot of the underneath of the undertray

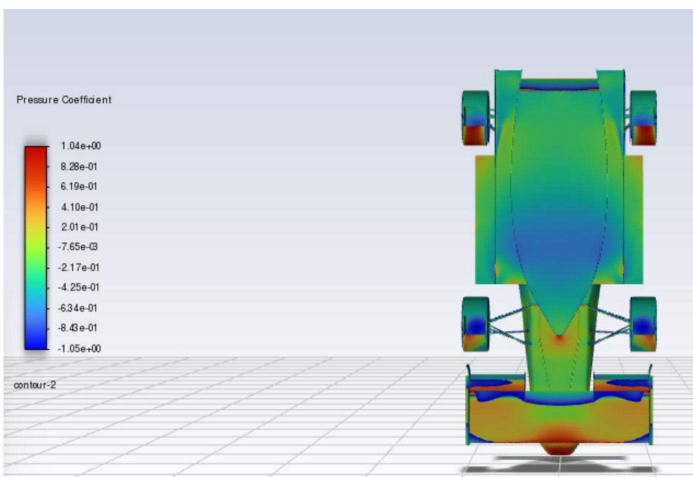


Figure 25: Pressure Coefficient surface plot of the underneath of the undertray and the car

Results @ 30 mph:

- Lift: -34 N
- Drag: 5 N
- L/D ratio: 7

Cooling

From our lap time simulations, we simulated that the accumulator, inverter and motor would produce combined heat of about 2500 W of heat during our endurance run and therefore decided to go with **water cooling** with cold plates installed in the accumulator and inverter. Here are more details:

- Pack total heat: 1600 W
- Cell heat produced: 3.7 W
- Total Area: 0.586 m^2
- Total Volume: 0.003 m^3

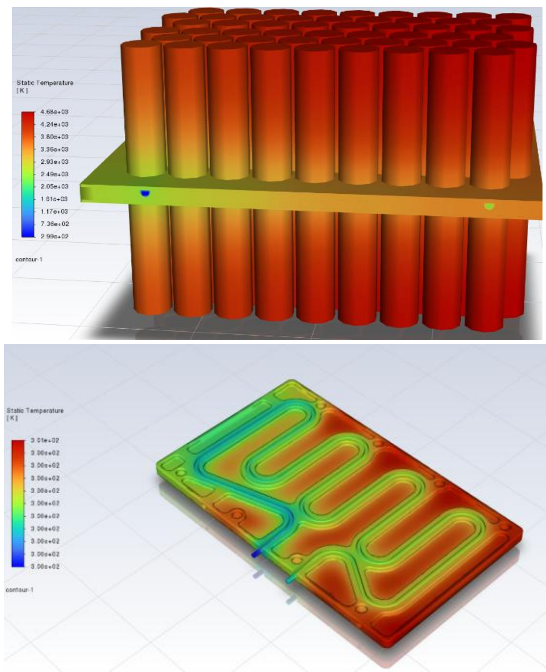


Figure 26: Static Temperature surface plot of the cells and coldplates

For effective heat transfer through the radiator, turbulent air is best for that. Therefore the best location to place the radiator was determined to be behind the driver on the left side. This location allows taking advantage of the turbulent air from the wake region of the front tire.

Fan parameters:

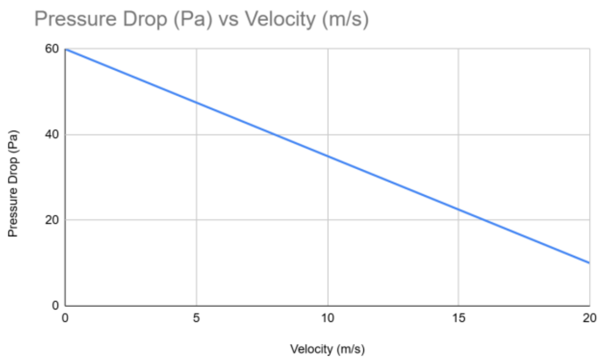


Figure 27: Pressure Drop vs Velocity through the fan

CFD @ 30 mph:

The radiator was simulated as a porous region with the following parameters:
- Face Permeability: 1,000,000
- Porous medium thickness: 0.04 m
- Pressure Jump Coefficient: 20.41 m^-1

Using the Darcy Forchheimer law which states that the pressure drop across a porous medium is given by:

$$\Delta P = \frac{\mu}{D}\upsilon + \frac{\rho}{F}\upsilon^2$$

The pressure drop across the radiator was calculated to be: *90.2 Pa*

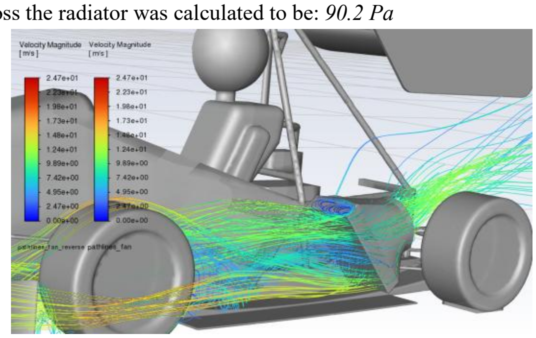


Figure 28: Velocity streamlines through the radiator

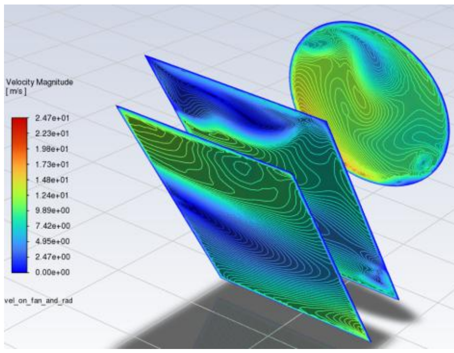


Figure 29: Velocity contours on the radiator and fan

Results:

- Mass flow rate through the radiator: 0.375 kg/s

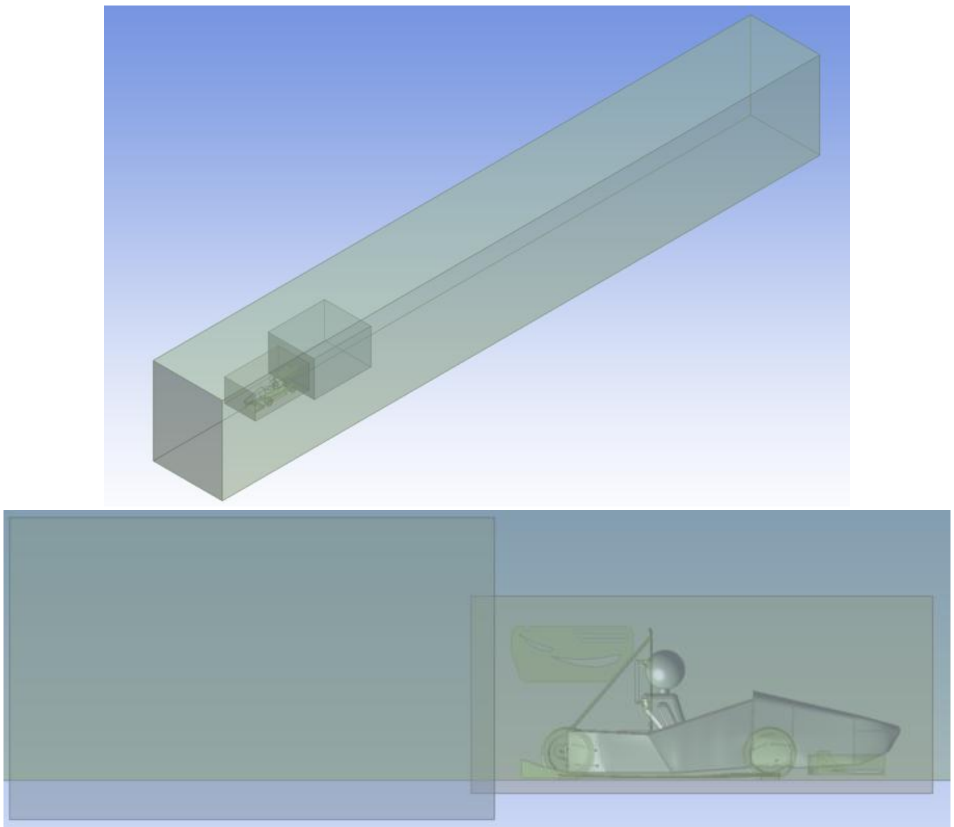
<u>Mass of the aerodynamic components</u>

Post-manufacturing, each aerodynamic component was weighed. Here are the results:

Component	Mass (lbs)	Downforce/Mass Ratio
Front Wing	7	38.8
Rear Wing	13	1.83
Undertray	6.2	1.2

Computational Fluid Dynamics (CFD) Model

To analyze and validate NFR 25’s aerodynamic design, the team ran CFD simulations on the car. This year, the team used Ansys Fluent as the preferred software. The simulations were set up as steady-state, and most of the base simulations were run at 30 MPH, as this was the average speed of the car during autocross. First, the team began with 2D simulations to visualize the initial designs and optimize them after examining the results. 2D simulations are not accurate representations of flow over a full car, so the team moved to a 3D analysis of the car after initial designs had been optimized.



Figures 30-31: Design Modeler Geometry

For the first initial runs, half-car simulations were done with symmetric conditions to decrease computational time. The turbulence model used for both 2D and 3D simulations is the k-omega SST model with a coupled scheme with second-order momentum and second-order upwind. The half-car 3D sims had around 9.4 million elements. An important part of the verification process for the team's CFD simulations is to ensure that the simulations converge. To do this, we check the residuals:

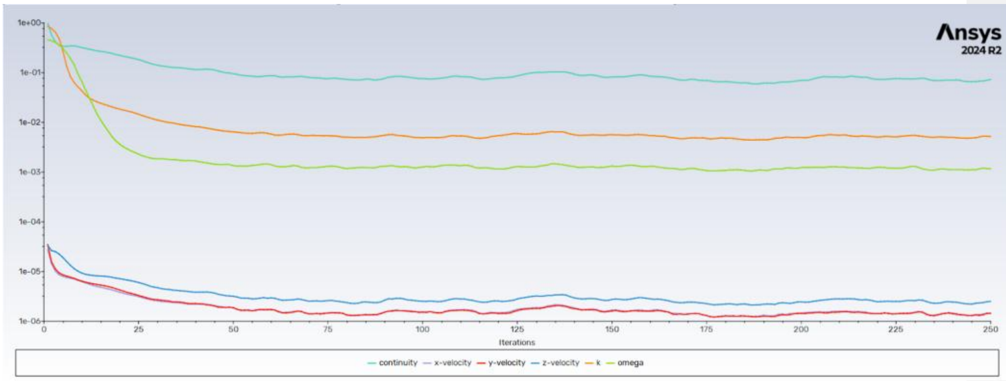


Figure 32: Residual Plot for Half-Car Simulation

In this plot, the residual plot does not fully converge. This can be due to the complex geometries, such as the rotating tires, the curvature of the wings, and the selected turbulence model. Using a k-epsilon model would help with the convergence because it is a more robust model, but it performs poorly in near-wall regions, and k-omega resolves the boundary layer more accurately. K-omega also blends into a k-epsilon in the freestream, which can help with solving flow separation. With this in mind, it is still important that the team can verify the sim and ensure there is some sort of convergence. Thus, the lift force result can be monitored to ensure that it converges to a certain value after many iterations.

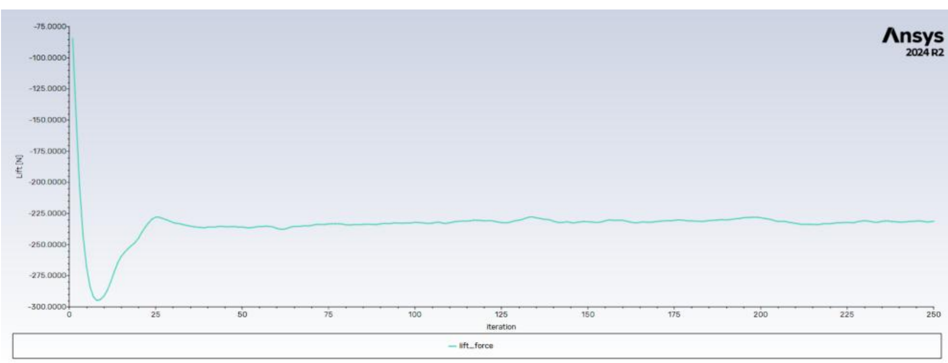


Figure 33: Lift Plot per iteration for Half-Car Simulation

Although not perfectly converged, this figure shows that the lift force stabilizes and has no major oscillations or divergence. Since the team is using a K-omega turbulence model, it is important to check the Y+ values for the walls.

Table 1: CFD Y+ Values

Wall	**Y+ Value**
Chassis, suspension, and accumulator	8.41
Front Wheels	10.61
Rear Wheels	9.43
Front Wing	4.29
Rear Wing	7.34
Undertray	5.87

Many of the walls have y+ values less than 10, which shows that the simulations are suitably meshed for most regions. With all of this now verified, the team ran many simulations with different iterations of our aerodynamic package. For the final design, the downforce is 193.10 N, the drag is 96.67 N, and the C_L/C_D is 2. These results show a good tradeoff between lap time (63.3s simulated endurance lap) and efficiency (0.4 Ah simulated remaining in battery after endurance run).

A crucial part of the team's design plan was to not just focus on generating as much downforce as possible, but to be able to have a center of pressure that is not too far from the center of gravity, which would provide the drivers with handling and grip issues. From our initial sims, our center of pressure was around 10 inches in front of the center of gravity. Thus, the team made some changes to the front wing to help decrease the distance between the CoP and the CoG. After some changes, the team ran more simulations and although the total downforce decreased slightly, the CoP was around 6 inches in front of the CoG.

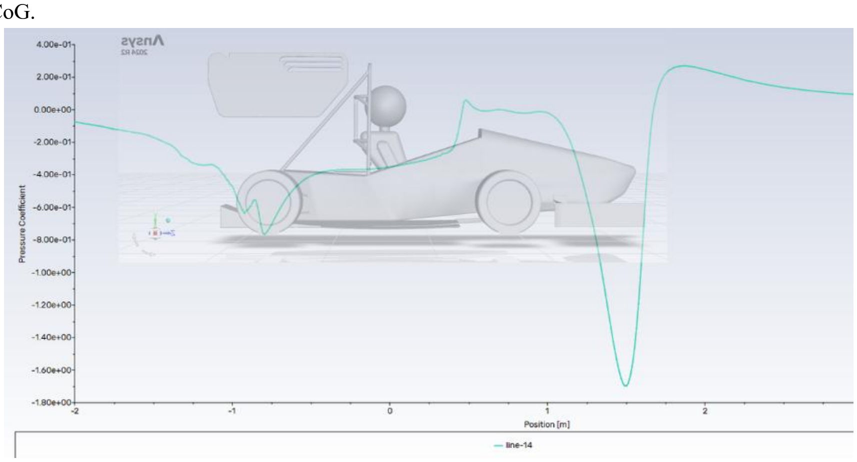


Figure 34: Plot of Pressure Coefficient Aligned with NFR25

The team also wanted to analyze the car during yaw, roll, pitch, different steering angles, and different ride heights. For the different roll simulations, it was observed that the sensitivity of the roll does not greatly affect the migration of the CoP. The team observed slight lateral migration, with a maximum of 2 cm at 1.5°. On the other hand, the pitch angle changes the downforce on the car. For the base case, the lift was 193 N, and as the car pitches upwards, the downforce decreases to 181 N (pitch angle of 0.8 ° upward). The CoP location also changes drastically. At 0.8 °, the CoP location is 1.18'' in front of the CoG and 3.54'' at 0.4 °. These results show that the car is much more stable when accelerating out of a corner. When the car pitches downwards, there is an increase in the downforce and CoP migrates toward the front tires. With a change in ride height, there is a slight change in the downforce generated on the car (see appendix), and the CoP does not migrate much, and stays at around the same location as the base simulation.

For all of the results and plots from the CFD simulations, please see the appendix.

On-Track Testing

After completing the simulations, the team wanted to compare and validate the results. To do this the team used driver feedback, yarn tufts, and pitot tubes.

The driver feedback from all of the test drivers were about the same. They mentioned that at low speeds the car felt more understeer and at higher speeds the car tended to feel more oversteer. This feedback tracks with what the team predicted as the difference between the location of the CoP and CoG will be felt at higher speeds and since our CoP further forward, oversteer can be expected. The team also wanted to get feedback from the drivers and compare the stability with and without the aerodynamic package. It was observed that NFR25's drivability does greatly improve when the aerodynamic package is on the car. Without the components, the car because much more unstable at high speeds and there is less grip.

The team then tested the aerodynamic package with aero tufts. These tufts are used to visualize the air flow, and observe any vortices or flow separation. A picture of the teams yarn setup as well as some pictures of the car in motion are shown below.

Figures 35-37: Yarn testing of aerodynamic components

In these pictures, the aligned tufts that can be seen on the nose cone, rear wing, and front wing show attached flow. There are some tufts on the rear wing's endplate that show some fluctuations which can indicate vortices being formed in those areas. The yarn on the outer edge of the front wing shows the yarn moving towards the right and towards the front tire, which is something that the team expected due to the interaction between the tire and the front wing. Overall, the tufts show that most of the flow is attached with vortices being created along the top edge of the endplates.

Another form of testing the team wanted to complete was pitot tube testing.

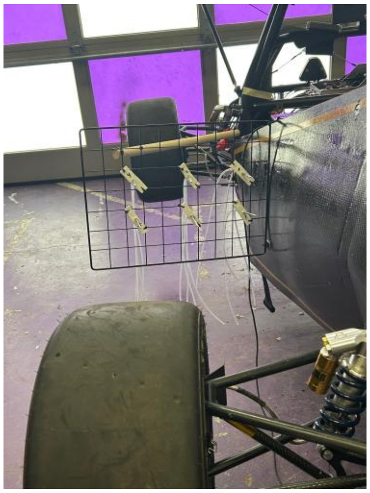


Figure 38: Pitot Tube Rake Setup

Above is a picture of the setup for the pitot tubes. There are six pitot tubes total, and each will measure the velocity of the air at the stagnation point. Knowing the velocity and the ambient pressure, the total pressure can be calculated:

$$V_\infty = \sqrt{\frac{2(P_0 - P_\infty)}{\rho}}$$

$$P_0 = \frac{{V_\infty}^2 \rho}{2} + P$$

After solving for this pressure at any point for the pitot tubes, and comparing these values with the ones we get from the CFD simulations.

Interactions

For this year's design, the main

Mechanical Design & Manufacturing

In order to make lightweight with high strength parts, carbon fiber was chosen as the ideal material to make the aerodynamic components. There are different manufacturing techniques used to make carbon fiber parts. The process that yields parts with the low weight is prepreg layups. Therefore, the front and rear wings were manufactured from oven cured prepreg carbon fiber. The undertray was manufactured using the wet layup technique since it could not fit in the oven.

Figure 39: Heat Gun Oven Curing

Figure 40: Front wing elements in the oven before curing

FDM Nylon 12CF 3D printed molds were used for the airfoil elements for the front wing and rear wing. Previously, our team relied on MDF and cardboard molds—options that were either time-intensive or prone to warping under the heat and pressure of high-temperature prepreg carbon fiber curing. These constraints limited not only our design freedom but also our confidence in the final product. With the move to 3D printed molds, we gained precision, repeatability, and reliability—cornerstones of high-performance engineering.

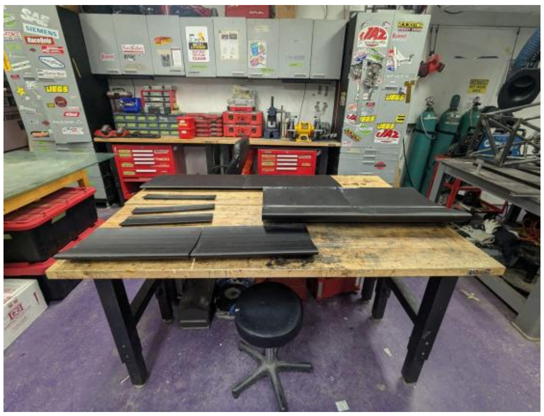


Figure 41: Front wing and Rear wing molds received from Stratasys

Five distinct aerodynamic elements across the NFR25's front and rear wings were manufactured using 3D printed molds. The transition brought immediate benefits. The robust thermal stability of Nylon 12CF—showing minimal dimensional change even at 120°C (~250°F)—meant molds held shape under cure cycles, producing accurate, high-quality parts without the defects we had previously seen.

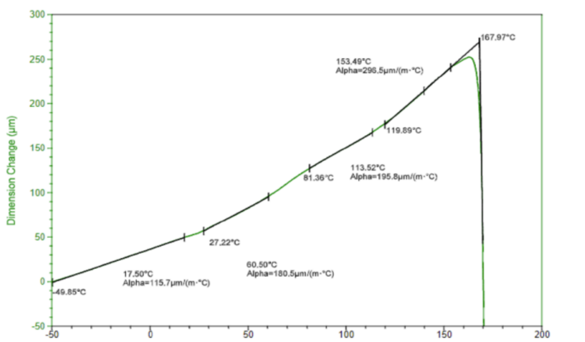


Figure 42: Dimension change data as a function of temperature for the Nylon 12CF Flat sample.

Figure 43: Rear Wing Main Foil Fully Cured.

Front wing mounting:

- 6061-T6 mounts ribs
- Mild steel frame tabs with multiple holes for front wing ground clearance adjustability
- Load:
  - Downforce: 61 N
  - Drag: 6 N
  - Weight: 22.24 N
  - Side Impact: 100 N
- Safety Factor: ~ 1.5
- Maximum deflection: 4 mm
- The mounts were designed to connect to ribs inside the hollow main foil
- The ribs were also made out o
- 10-32 Fasters: 150 ksi

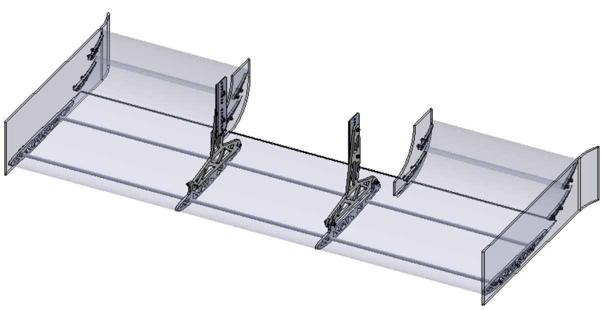


Figure 44: Front Wing Mounting system

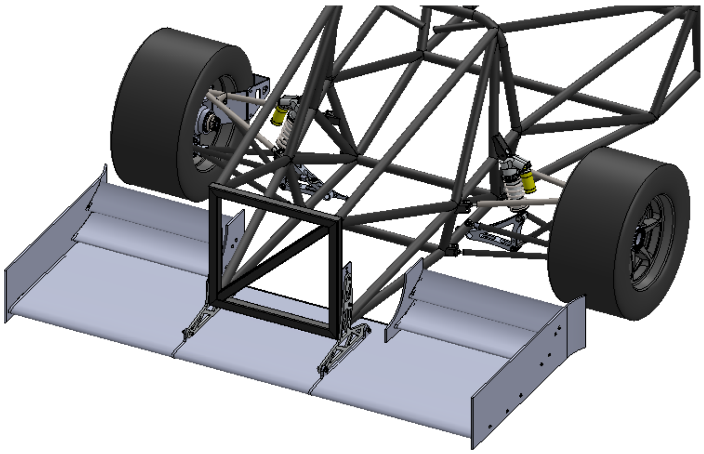


Figure 45: Packaging of the Front Wing with the front tires and the frame

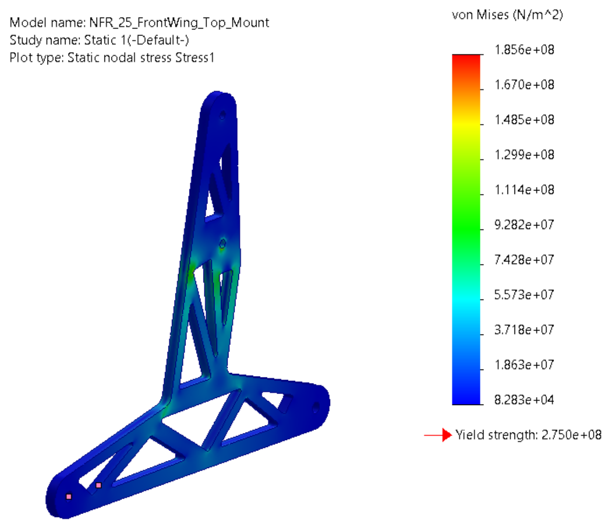


Figure 46: Stress plot of the Front Wing mount

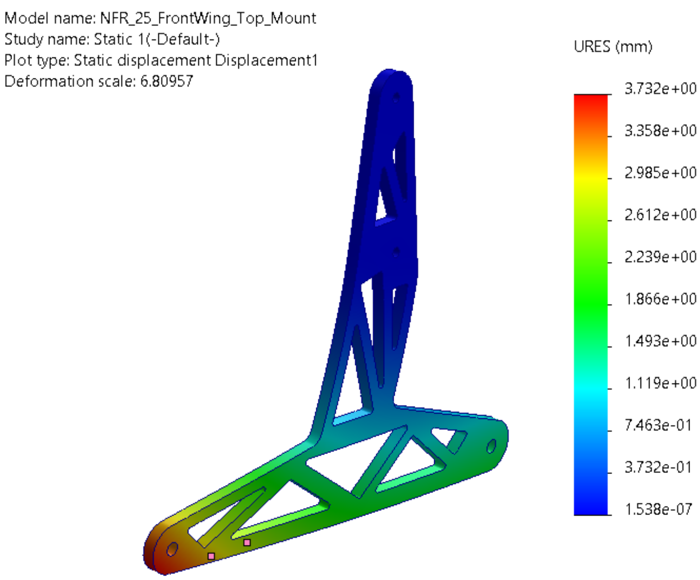


Figure 47: Deformation plot of the front wing mount

Rear wing mounting:

- 6061-T6 swan neck and ribs attached to the Main Hoop and Main Hoop Bracing Node
- Carbon Fiber arms

Figure 48: Mounting system of the rear wing

- Load calculations:

  Coordinate of the leading edge of the mainfoil : (0, 35.24, -16.06) in
  Therefore to get the center of pressure, use y = 35.24 in ~ 0.8952 m

  Center of mass of rear wing: (0.00 , 36.38, -28.15) in

  Center of pressure location: (-0.276, 0.895, -0.806) m

  Due to half a car, X is shifted to the right(in direction of the driver forward) i.e. x_cop = 0

  CoP: (0.0, 0.895, -0.806) m = (0, 35.24, -31.73) in

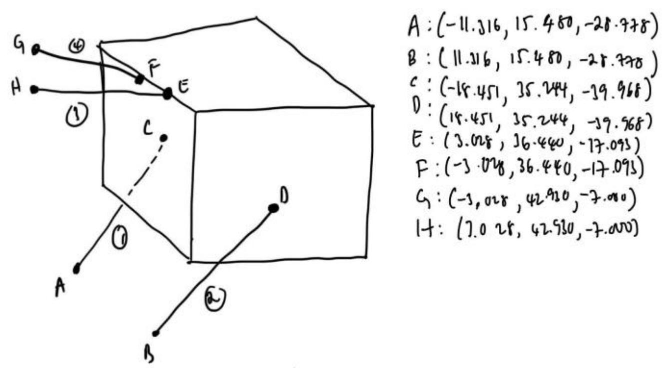


Figure 49: Chassis Attachment points for the rear wing mounting

  Center of side force: (19.75, 38.28, -29.30) in

  Results:
  - Swan neck: Max load 398.6 lbf vertically, -138.1 lbf horizontally from static load
    - Safety Factor: ~ 1.8
    - Maximum deflection: 0.6 mm

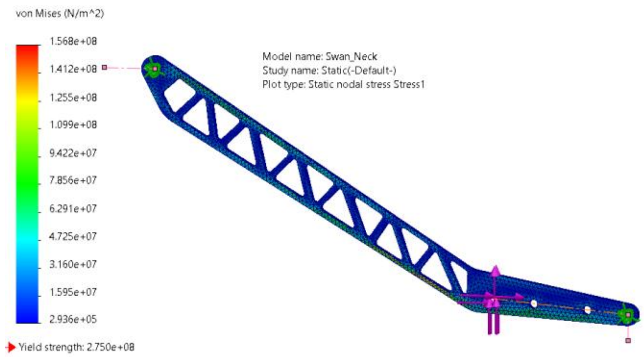


Figure 50: Stress plot of the Swan Neck

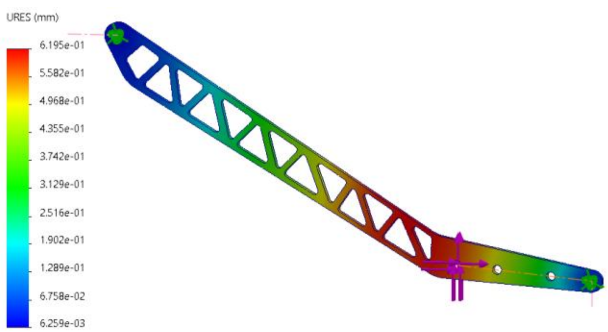


Figure 51: Deformation plot of the Swan Neck

- Carbon fiber arms (0.5” ID, 0.625” OD): Max axial load 97.6 lbf at 60 mph no cornering
    - Compressive strength: 700 ksi
    - Cross Sectional area: 0.110 in^2
    - Axial stress: 887 psi
    - Safety Factor: Compressive Strength >>> Axial Stress

Undertray mounting:
- ¼-28 fasteners
- Four attachment points on the chassis: two on running tubes and two on the rear bulkhead

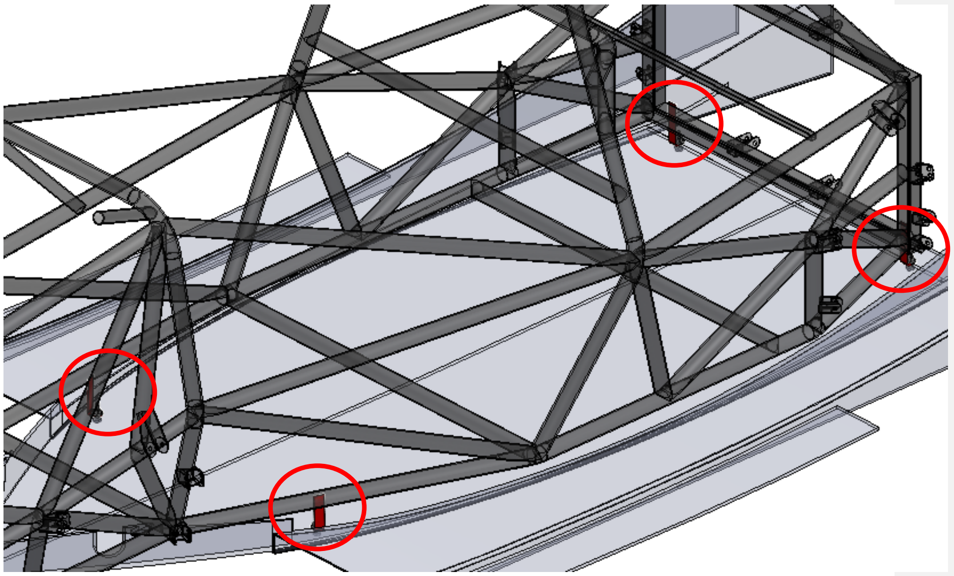
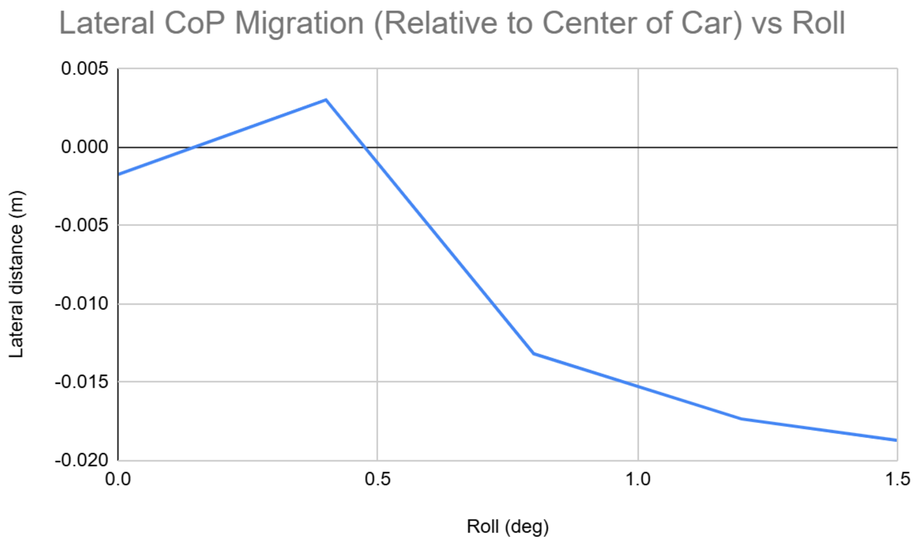
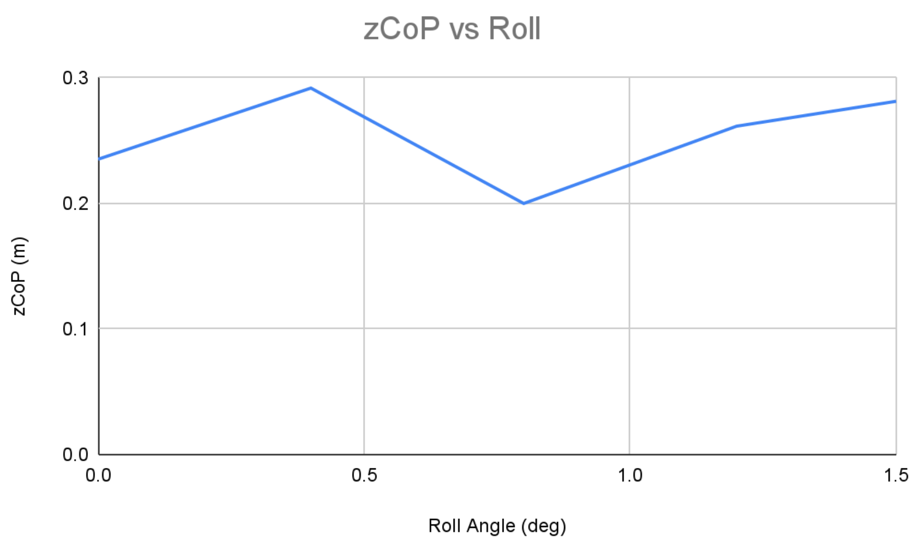
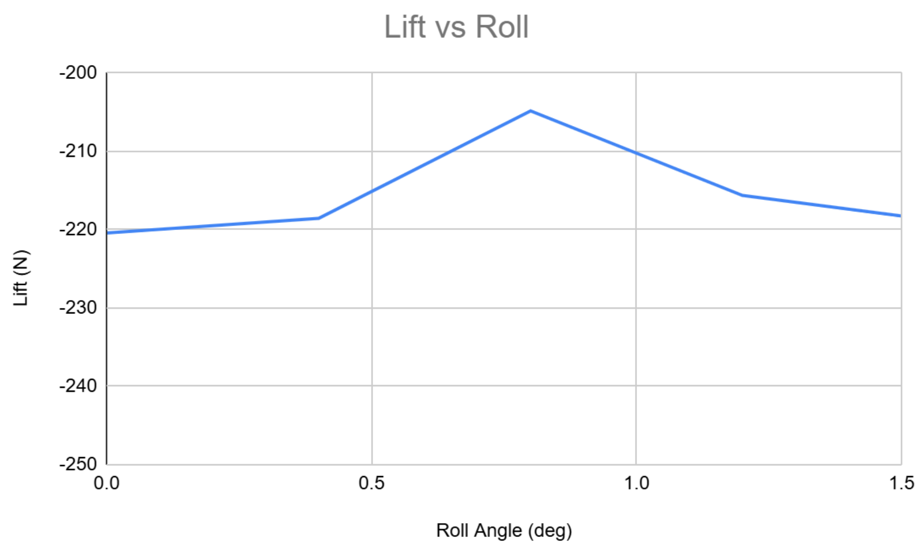


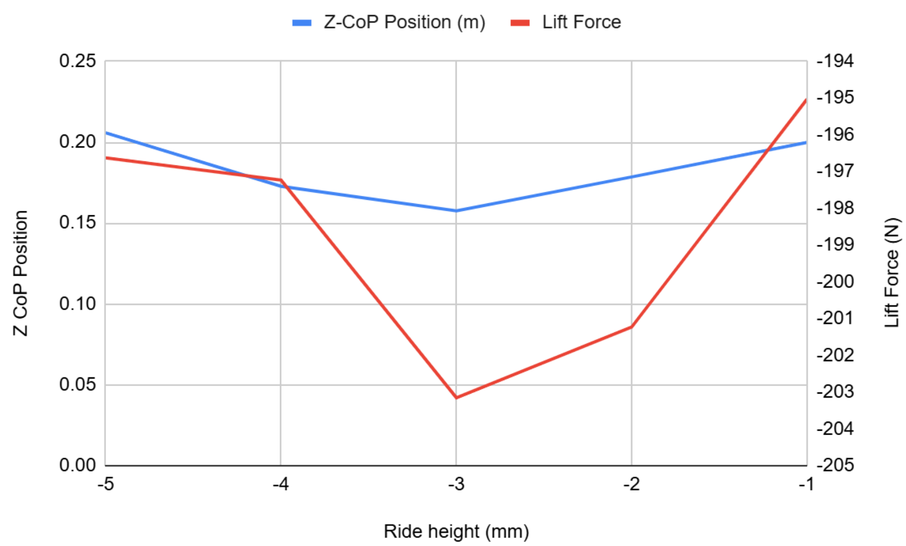
Figure 52: Undertray chassis attachment points

Appendix A: CFD & Mesh Sensitivity

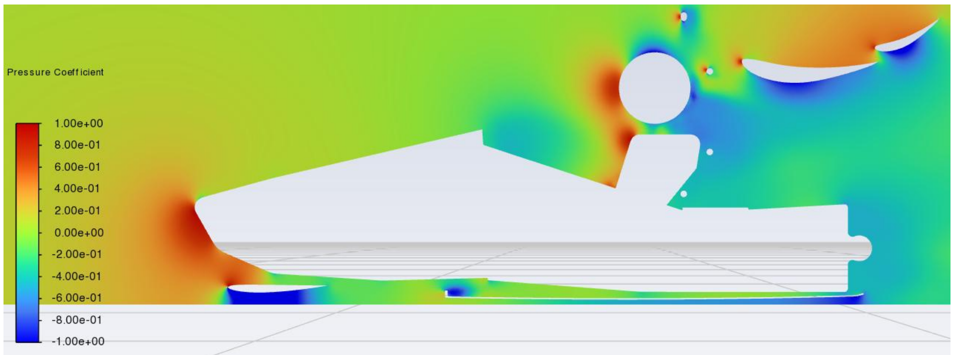




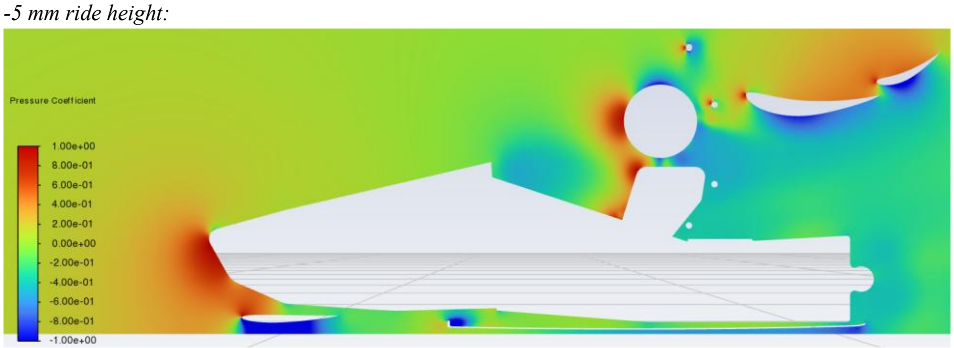


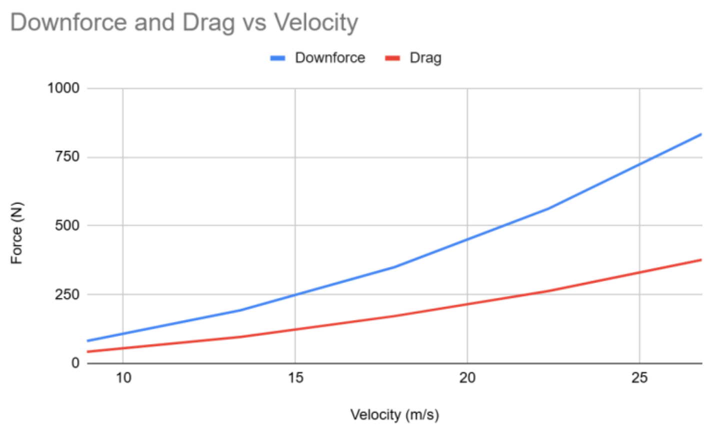


-3 mm ride height:

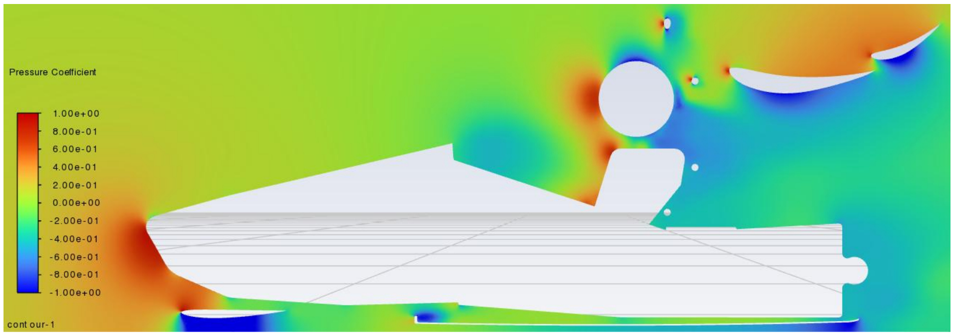


-5 mm ride height:

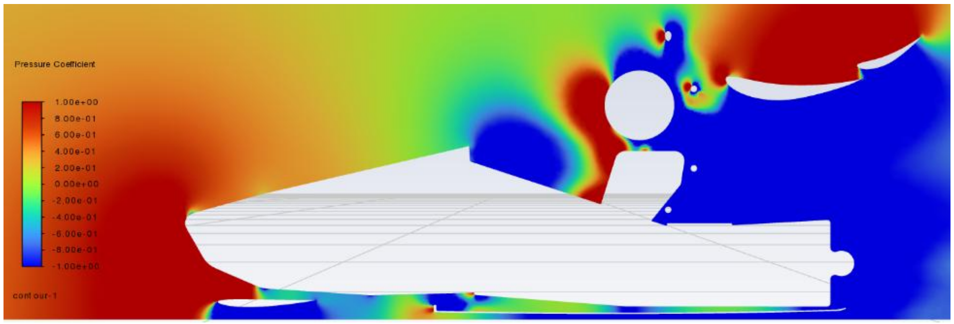


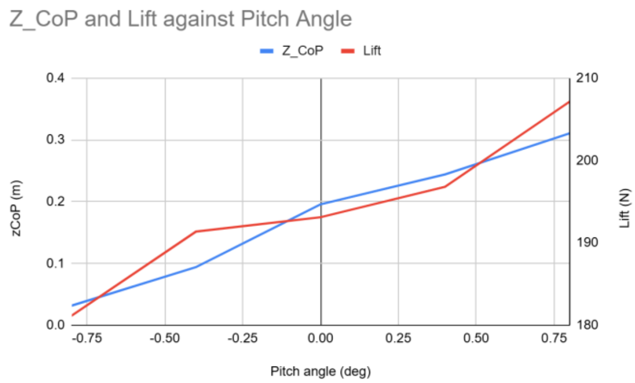

Downforce and Drag vs Velocity
Downforce
Drag
1000
750
500
250
0
Force (N)
10
15
20
25
Velocity (m/s)

30 mph:

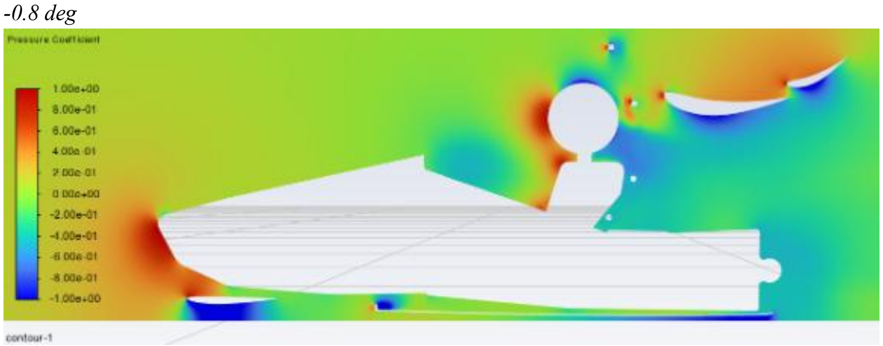

Pressure Coefficient
1.00e+00
8.00e-01
6.00e-01
4.00e-01
2.00e-01
0.00e+00
-2.00e-01
-4.00e-01
-6.00e-01
-8.00e-01
-1.00e+00
contour-1

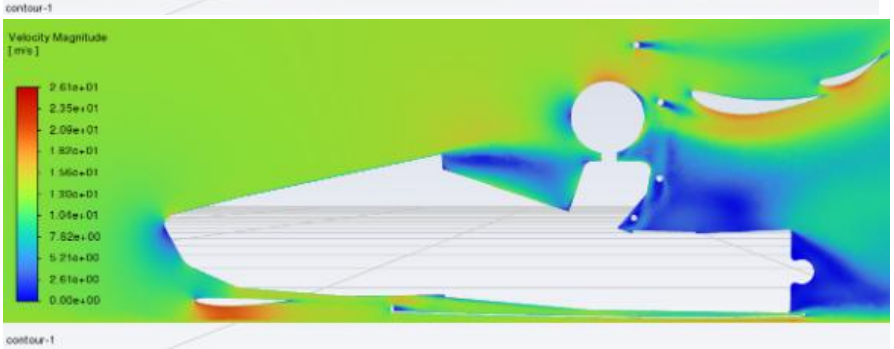
60 mph:


Pressure Coefficient
1.00e+00
8.00e-01
6.00e-01
4.00e-01
2.00e-01
0.00e+00
-2.00e-01
-4.00e-01
-6.00e-01
-8.00e-01
-1.00e+00
contour-1

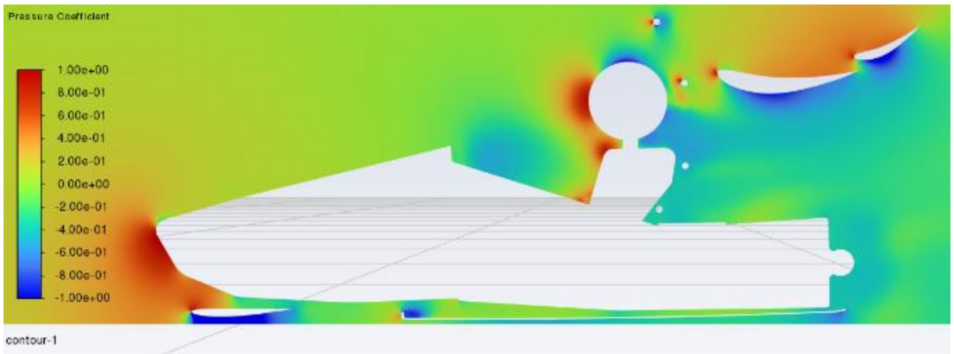

Z_CoP and Lift against Pitch Angle
Z_CoP
Lift
zCoP (m)
Lift (N)
Pitch angle (deg)
0.0
0.1
0.2
0.3
0.4
180
190
200
210
-0.75
-0.50
-0.25
0.00
0.25
0.50
0.75

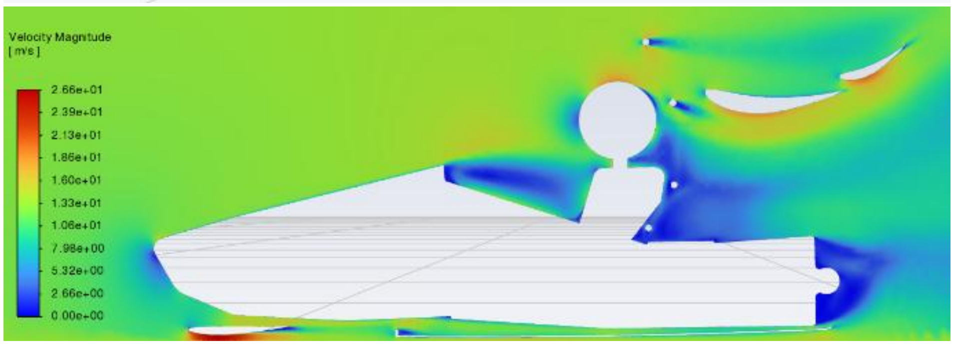
-0.8 deg

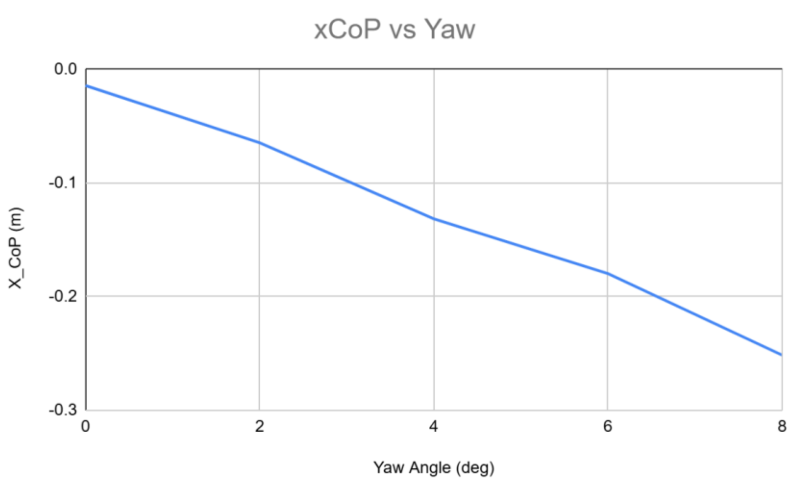

Pressure Coefficient
1.00e+00
8.00e-01
6.00e-01
4.00e-01
2.00e-01
0.00e+00
-2.00e-01
-4.00e-01
-6.00e-01
-8.00e-01
-1.00e+00
contour-1

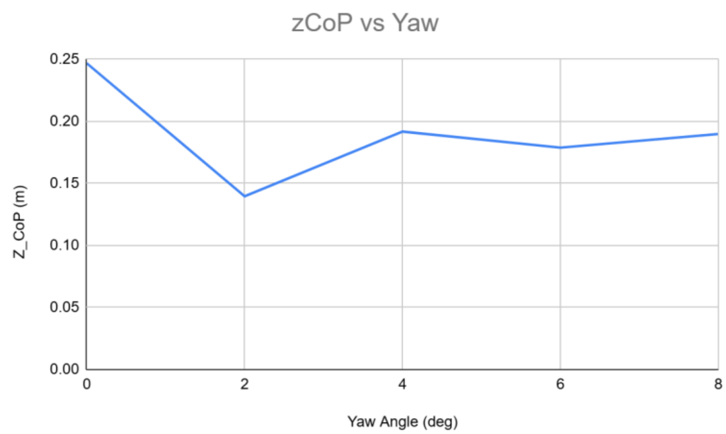

Velocity Magnitude
[m/s]
2.61e+01
2.35e+01
2.09e+01
1.82e+01
1.56e+01
1.30e+01
1.04e+01
7.82e+00
5.21e+00
2.61e+00
0.00e+00
contour-1

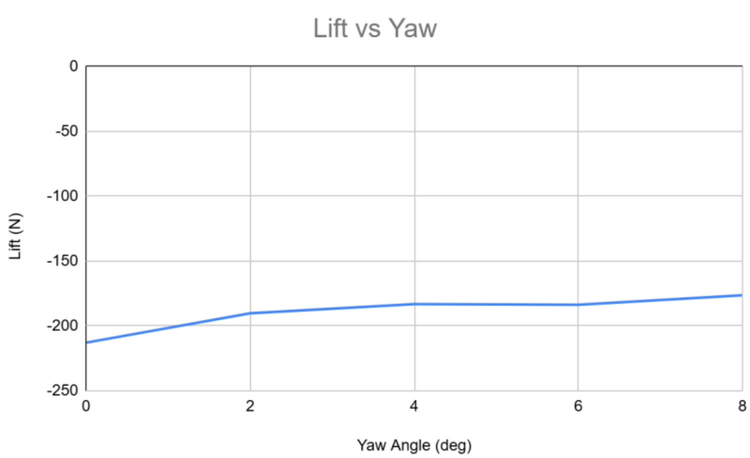
0.8 deg


Pressure Coefficient
1.00e+00
8.00e-01
6.00e-01
4.00e-01
2.00e-01
0.00e+00
-2.00e-01
-4.00e-01
-6.00e-01
-8.00e-01
-1.00e+00
contour-1


Velocity Magnitude
[m/s]
2.66e+01
2.39e+01
2.13e+01
1.86e+01
1.60e+01
1.33e+01
1.06e+01
7.98e+00
5.32e+00
2.66e+00
0.00e+00


xCoP vs Yaw
0.0
-0.1
-0.2
-0.3
X_CoP (m)
0
2
4
6
8
Yaw Angle (deg)


zCoP vs Yaw
Z_CoP (m)
0.25
0.20
0.15
0.10
0.05
0.00
0
2
4
6
8
Yaw Angle (deg)


Lift vs Yaw
Lift (N)
0
-50
-100
-150
-200
-250
0
2
4
6
8
Yaw Angle (deg)